Tram speed and event recorder under wheel slide: engineering dossier

Rev A, 11 Nov 2026. Portfolio and thesis-preparation work; not a product, not certified.

This dossier follows the systems-engineering spine of Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed. (Wiley, 2016) [W]. Each section names the chapter and pages it applies. Every engineering number carries a reference key [Rn] (list in §17, details in docs/sources.md) and a status: - **V**: read in the source. **V2**: read in a secondary source that quotes the primary. - **D**: derived from V values by a stated method, or computed by the models of this repository. - **A**: an assumption of this study (all listed in §14).

The models, firmware and tests that produce the results are in this repository. Study IDs (S1–S7) refer to `model/studies/`, test IDs (ITxx) to `fw/tests/robot/odo.robot`, requirement IDs (OR-xx) to `fw/SRS.md`.

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0. The question and the short answer

Open engineering question. A tram speed and event recorder is specified to keep the last 1500 m of travel "with 0.25 m accuracy" and to measure the brake-test stopping distance "with 5 cm accuracy", and to keep its record when the vehicle supply is cut [R19] (V). In an emergency brake on a wet or leaf-covered rail the wheels slide, and on a modern low-floor tram every axle is braked, so no wheel rolls freely. **Can the recorder still measure the stopping distance, and what sensor set, estimator and hold-up energy does it take?**

Short answer, from the models and firmware in this repository:

1. **Wheel pulses alone cannot measure an emergency brake on a wet or leaf-covered rail** (S3, §9.3). - With one wheel sensor (today's recorder, E1) the stopping distance is under-read by a median of 0.86 m and 4.8 m at the 95th percentile over 192 simulated brakes; on leaves the p95 is 5.2–5.3 m. - The cause is not only the slide. At the wheel-brake demand, the Polach creep alone is 0.19 % on a dry rail and 2.0 % on a wet one (S1), so even a wheel that does not slide reads 2 % short on a wet rail. - A second wheel sensor on another bogie (E2, larger increment) helps little: p95 2.9 m.
2. **Two wheel sensors plus a 6-axis MEMS IMU, with the wheels switched out for the whole brake, meet the brake requirement where it is defined** (E3, the chosen design). - Over the full envelope (dry, wet, dry and wet leaves; gradients to ±8 %; vertical curves; 38–52 t; 30–70 km/h; wheel wear 630–680 mm): median error **3.0 cm**, p95 **14.7 cm**, worst 28.6 cm, i.e. 0.3 % of the stopping distance at p95. - In the depot brake test (dry, level, empty, 30–50 km/h, calibrated wheels) the p95 is **2.5 cm** and every run is inside 5 cm (S4). - The distance comes from the accelerometer integral, started at the wheel speed before the brake and closed by the condition that the speed is zero at the detected standstill (two-point boundary correction, §8.3). - The **pitch gyro is necessary**. Without it (accelerometer only, E3a) the p95 is 1.9 m: on a vertical curve or under the body's pitch on its suspension, gravity along the car body changes during the brake, and the accelerometer cannot tell it from deceleration. - A **Doppler radar** (E4) gives p95 27 cm in the brake, but over 1500 m its ±0.5 % scale error [R10] makes 7.9 m (p95). Adding it to E3 (E5) does not help the brake (p95 30 cm).
3. **"0.25 m over the last 1500 m" can only be the sampling step, not an accuracy.** - 0.25 m in 1500 m is 0.017 %. A wheel diameter known to 0.05 % gives 0.75 m at the far end of the window; the best estimator here (E3) reaches 3.0 m p95 there, because traction creep and calibration act on the whole window (S3, S4). The firmware records one point every 0.25 m (OR-30) and reports the error budget instead of claiming a 0.25 m accuracy.
4. **The brake onset must be time-stamped, not sampled.** A 10 ms input sample is 19 cm at 70 km/h, four times the brake-test tolerance. The firmware integrates only the part of the onset tick after the captured edge (OR-20).
5. **Keep the record in FRAM, not in RAM flushed to an SD card** (S5, §7.4). - With every record written to FRAM when it is made, 353 µF on the 24 V input carry the unit through an EN 50155 class S2 interruption (10 ms) [R4] and finish the record; the power-fail record is in FRAM 16 µs after the comparator edge (Renode, IT05). - Flushing a RAM ring of 1500 m (96 KB) to an SD card can take 1.25 s at the SD busy limit [R14]: that needs 0.53 F of supercapacitor.
6. **Architecture.** In the analysis of alternatives the two wheel sensors plus IMU (A3) score 7.94 against 6.46 for the one-sensor recorder; A3 wins 70 % of 20 000 random weightings and no ±10 % change of a single weight changes the winner (S6, §8.1).

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1. Problem space (W ch. 4, pp. 76–98)

Problem statement (one sentence, no solution words; W p. 92). After a tram collision or a brake test, the operator and the investigator cannot trust the recorded distance and speed of the last metres when the wheels were sliding.

Discussion. - An urban tram must stop from 70 km/h within 69 m in an emergency brake (mean deceleration at least 2.73 m/s²) on dry level track, and from 50 km/h within 39 m [R1 Table 2] (V). To get there it uses magnetic track brakes, which do not depend on adhesion, and wheel brakes, which do. - Wheel-rail adhesion falls from about 0.25–0.45 on a dry rail to about 0.25 on a wet one [R2] and to 0.04 on a wet leaf layer in field tests (dry leaf layers 0.06–0.11) [R3] (V). Under emergency braking on such a rail the wheel slide protection releases and re-applies each axle; the wheels turn slower than the vehicle moves. - Recorders of driving data exist for accident investigation on all rail vehicle types [R7] (V2). A published tram recorder keeps the last 1500 m and measures the brake test from the wheel pulses, with the wheel diameter entered as a parameter [R19] (V). - On a tram all axles are motored or braked (A). So the classic odometry trick of a free-rolling reference axle is not available.

Need typing (W Principle 4.12, p. 89). - **Real need:** a stopping distance and a speed profile that hold up in an investigation, also when the wheels slid. - **Projected need:** the same record as input to driver-assistance and collision-warning functions. - **Perceived need:** "more resolution" (0.25 m, 5 cm). Resolution is not accuracy (§0 item 3).

Stakeholders (after W Table 4.2, pp. 84–85).

Role	Description	Recorder concern
Driver (user)	operates the tram	the record must not blame the driver for the wheels' slide
Operator's depot (maintainer)	brake tests, wheel turning	5 cm brake test; diameter entry after wheel turning
Accident investigator (end user)	reconstructs the last 1500 m	known error bounds; integrity after a supply cut
Vehicle integrator	supply, mounting, cabling	24 V supply per EN 50155; two sensor cables; one box
Safety assessor	software and hardware evidence	EN 50716 / EN 50657 process [R5], EN 50126/50129 [R6]
Authority (BOStrab)	braking performance	Anlage 2 limits [R1]

Problem and solution spaces (W Principles 4.17–4.18, Fig. 4.7, pp. 93–95). Problem 1.0 "know where the tram was and how fast" splits into 1.1 sensing (wheel, inertial, ground speed), 1.2 estimation, 1.3 recording and integrity, 1.4 power, 1.5 calibration. The question of §0 is a **critical technical issue** (W p. 100): the estimator, the sensor set and the hold-up energy must be chosen together.

2. Mission analysis, use cases and scenarios (W ch. 5, pp. 99–128)

Mission statement (one outcome, W Principle 5.2): every metre and every brake of the tram's service is on record with a known error, and the record survives the end of the supply.

Mission objectives. - **Performance:** stopping distance of an emergency brake to 5 cm in the depot brake test and to 0.30 m (p95) in service, including slide (§5). - **Integrity:** no record lost or silently wrong after a supply cut. - **Constraint:** one box in the cab, 24 V supply per EN 50155, sensors that a depot can maintain.

Phases (W Principle 5.7, p. 110): pre-mission (power-up, journal recovery, standstill calibration of the IMU offsets at the first stop); mission (runs between stops, service and emergency brakes); post-mission (record download, wheel turning and diameter calibration).

Mission event timeline (W Fig. 5.5, p. 112):

```
t0 power on -> t1 BOOT record, journal recovered -> t2 first standstill: IMU offsets learnt
-> t3 departure (traction creep) -> t4 cruise on gradients -> t5 emergency brake (slide, track brakes)
-> t6 standstill: corrected BRAKE_RES record -> t7 supply cut: PFAIL record, last record in FRAM
```

Use cases (W Table 5.1, p. 123; Principle 5.23, pp. 122–125).

UC	Actor	Normal outcome	Abnormal outcome (W Fig. 19.5)
UC1 emergency brake in service	driver, investigator	BRAKE_ON and BRAKE_RES records, distance within 0.30 m p95	wheels slide: IMU carries the distance (E3)
UC2 depot brake test	maintainer	distance within 5 cm	uncalibrated diameter: CALBAD flag in every record
UC3 accident reconstruction	investigator	last 1500 m at 0.25 m spacing with flags	torn last record: recovery keeps all complete ones
UC4 wheel turning, calibration	maintainer	diameters from two marks 100 m apart	result outside 0.600–0.700 m rejected per axle
UC5 supply interruption	vehicle	full function through 10 ms (S2)	supply gone: PFAIL record, journal consistent
UC6 sensor failure	—	degraded mode, flagged	both wheel sensors failed: brake event stays open

3. Concept of operations (W ch. 6, pp. 129–146)

- The recorder is a cab box. It takes two wheel pulse generators (one axle on bogie 1 and one on bogie 3, two channels, 90°), a 6-axis IMU on its own board, the emergency-brake demand and a calibration-mark input as binary inputs, and the power-fail comparator of its own supply (after W Table 6.4, p. 143).
- At every stop it learns the accelerometer and gyro offsets (the tram stands still: speed 0, specific force = offset, pitch rate 0).
- While moving it estimates speed from the wheels, checked against the IMU; a wheel that disagrees is gated out (slide, spin, sensor fault).
- When the emergency-brake demand rises it stops using the wheels, integrates the IMU from the speed it had at the captured edge, and at the standstill closes the integral with the condition v = 0.
- It writes a record every 0.25 m and at every event to FRAM; an SD card (not modelled) keeps the long-term archive, copied in the background.

4. Phases, modes and states (W ch. 7, pp. 147–173)

States (W Table 7.1 format, p. 158; transitions as in W Fig. 7.14, p. 171):

State	Entry	Behaviour	Exit
INIT	power-up	journal recovery, configuration check	first sensor sample (OR-01)
STAND	every healthy wheel at rest	standstill updates of speed, offset and gyro offset (OR-13)	wheel moves
RUN	a wheel moves	wheel updates through the gate (OR-14), records every 0.25 m	rest; brake edge
BRAKE	brake-demand rising edge (OR-20)	no wheel updates; inertial distance (OR-21)	standstill confirmed (OR-22)
CAL	calibration command at standstill (OR-40)	counts at two mark edges	second mark (OR-41)

Overlay conditions (flags in every record): PFAIL (supply lost, unit still running on the hold-up), WHEEL0/WHEEL1 (sensor failed, OR-17), IMU (latched fault, wheels only, OR-50), SLIDE (a wheel gated out while moving), CORR (brake result corrected), CALBAD (default diameter in use, OR-42).

Degraded modes: one wheel sensor failed → estimator on the other; IMU failed → wheel-only brake without correction (OR-53); both wheel sensors failed → no standstill evidence, the brake event stays open and is recorded as such (FMEA W2, §10).

5. Requirements (W ch. 19–22, pp. 397–464)

System requirements derived from the sources (statement form after W Principle 22.5, p. 449; each with a verification method after W Table 20.1, pp. 417–418). The software requirements OR-xx in `fw/SRS.md` derive from these.

ID	Requirement	Source and derivation	Verification
SYS-1	The recorder shall record speed, distance and status continuously from power-up, on a 24 V supply from 16.8 to 36 V	[R19] (V); EN 50155 0.7–1.25 Un continuous [R4] (V2)	inspection; IT01
SYS-2	The recorder shall record the last 1500 m at 0.25 m spacing, with a distance error over the window within the calibration budget (0.05 % of the distance + 0.5 m)	[R19] (V) for 1500 m / 0.25 m; the accuracy is reinterpreted (§0 item 3, D)	analysis S3, S4
SYS-3	The recorder shall measure the emergency-brake stopping distance to 5 cm in the depot brake test (dry, level) and to 0.30 m (95th percentile) in service including wheel slide	[R19] (V) for 5 cm; 0.30 m = 0.4–3 % of the BOStrab stopping distances 9–69 m [R1] (D)	S3, S4; IT02–IT04
SYS-4	Every record shall carry its status flags and an integrity check	purpose of a driving-data recorder [R7] (V2)	test OR-25, OR-31
SYS-5	The recorder shall keep full function through a 10 ms supply interruption and keep every complete record after a supply cut	EN 50155 class S2 [R4] (V2); [R19] (V)	S5; IT05, IT06
SYS-6	The recorder shall process a sample in less than 10 % of the sample period	margin (A)	IT08
SYS-7	The recorder shall detect a failed wheel sensor or IMU and continue in a flagged degraded mode	FMEA §10 (D)	tests OR-17, OR-50, OR-53
SYS-8	The recorder shall accept sensor samples and commands and report status over a checked serial frame	interface (A)	tests OR-60..62
SYS-9	The software shall be developed with requirements traceability, structural coverage and a coding standard	EN 50657 / EN 50716 [R5] (V2): SIL-dependent techniques	CI evidence §12

Abnormal and emergency operations (W Fig. 19.5 and Principle 19.22, pp. 408–411) are explicit requirements (OR-17, OR-22 fallback, OR-33, OR-35, OR-50, OR-53), not left to the design.

Requirement validation (W Principle 20.1, p. 418, and 20.5, p. 424): SYS-3's 5 cm is kept only where it is achievable and meaningful (the depot brake test); in service the 0.30 m p95 is derived from the stopping distances and verified by the Monte Carlo. SYS-2 changes the source's wording on evidence (S4).

6. Architecture and interfaces (W ch. 26–27, pp. 542–598)

Physical architecture (W Fig. 26.9, pp. 562–564):

```
bogie 1 axle 1 --[2-ch Hall, 100 teeth]--+
bogie 3 axle 1 --[2-ch Hall, 100 teeth]--+--> QTMR/ENC quadrature capture --+
IMU board (accelerometer + pitch gyro) ------ SPI ---------------------------+--> i.MX RT1064 (Zephyr)
brake-demand input, mark input --------- opto inputs, edge capture ----------+     lib/odo: estimator,
24 V supply -> reverse diode -> 1000 uF hold-up -> DC/DC -> 3.3 V                 brake, journal
           \-> power-fail comparator ------------------------------------------>   GPIO interrupt
                                                  SPI FRAM 1 Mbit (record ring) <-+
                                                  SD card (archive, background)   <-+
                                                  vehicle bus / display (status)  <-+
```

Interfaces (keyed source–destination, W Figs. 27.4/27.5, p. 590; W Principle 27.6):

ID	From → to	Content	Rate	Check
I1	wheel sensors → capture	quadrature edges, 400 per revolution	up to 3.8 kHz at 70 km/h (D)	stuck-sensor check OR-17
I2	IMU → MCU	specific force (x), pitch rate (y)	100 Hz, 50 Hz filter	range check OR-50
I3	brake demand → capture	rising edge, µs time	event	onset OR-20
I4	comparator → GPIO interrupt	supply below threshold	event	PFAIL OR-35
I5	MCU → FRAM	16-byte records, CRC-16	≤ 4 per 10 ms	recovery OR-32/33
I6	MCU ↔ vehicle/rig	frames A5 5A len type payload CRC16	100 Hz in, 10 Hz out	OR-60..62

In the emulator I1–I3 are carried by the rig's sensor frame (the sampled front end): Renode's RT1064 model has neither the quadrature decoder nor the IMU (§14).

7. Performance budgets and margins (W ch. 31, pp. 666–681)

7.1 Stopping-distance error budget (S4; method of W Fig. 31.1 and Example 31.3, pp. 669–671)

Each error source alone, at its data-sheet value, on the same 24 brakes of the full envelope (p95):

Source	Value	p95 error	Note
model and algorithm alone (ideal sensors)	—	9.5 cm	onset ramp, standstill detection, traction creep in v0
wheel-diameter calibration	±0.05 % (A)	9.6 cm	enters through v0 only
accelerometer offset	±75 mg [R9]	9.5 cm	learnt at standstill and through the wheels
accelerometer scale	0.1 % + 0.01 %/°C × 60 °C [R9]	11.7 cm	the brake changes f by ~3 m/s²
accelerometer noise	25 µg/√Hz [R9]	9.4 cm	negligible
car-body vibration	0.1 m/s² rms, 1–10 Hz (A)	8.8 cm	band-limited: does not integrate into drift
gyro offset and drift	±1 °/s, ±0.015 °/s/K × 2 K [R20]	14.7 cm	the dominant sensor term ($g\,\delta b\,T^3/12$)
gyro noise	0.014 °/s/√Hz [R20]	11.1 cm	
gyro g-sensitivity, 10 % left after calibration	0.01 °/s/g (A on [R20])	9.2 cm	uncompensated (0.1 °/s/g): median doubles, 2.4 → 4.7 cm
all together		**14.6 cm**	budget SYS-3: 30 cm p95, margin 2×

The terms do not add: most of them are absorbed by the boundary correction. The residual is set by what changes **during** the brake: the gyro offset error grows linearly in time and leaves $g\,\delta b\,T^3/12$ after the correction (§8.3).

7.2 Calibration and the 1500 m window (S4)

Diameter error	Brake p95 (all conditions)	Error at the far end of 1500 m
0.01 %	15.5 cm	0.15 m
0.05 %	14.6 cm	0.75 m
0.1 %	13.7 cm	1.5 m
0.2 %	14.0 cm	3.0 m

0.5 %	25.6 cm	7.5 m

The brake distance is insensitive to the diameter (the wheels only give v0); the 1500 m window is proportional to it. The calibration over two marks 100 m apart at 1 m/s resolves 1 cm in 100 m (0.01 %, OR-41); IT07 measured the two diameters to −18 µm and +4 µm (0.003 %). Wear between calibrations (680 → 630 mm over the wheel life [R11]) is the reason to recalibrate after every wheel turning and at a fixed mileage (A).

7.3 Timing budget

Item	Budget	Measured
one sample (wheel update, filter, brake, records)	1 ms (10 % of 10 ms, SYS-6)	63 µs worst case (IT08, Renode virtual time)
power-fail record after the comparator edge	before the hold-up is exhausted (≥ 10 ms)	16 µs (IT05)
brake onset resolution	5 cm / 19.4 m/s = 2.6 ms (D)	captured edge, µs (OR-20)

7.4 Memory and hold-up energy (S5)

- Ring: 1500 m / 0.25 m = 6000 records × 16 B = 96 KB; 8192 slots = 128 KB = one 1-Mbit FRAM (D).
- Energy: $E = P\,t_{ride} + P_{finish}\,t_{finish}$; capacitor $C = 2E / (\eta\,(u1^2 - u2^2))$ from 16.8 V (0.7 Un [R4]) down to 9 V (DC/DC minimum, A), $\eta = 0.85$, $P = 3$ W (A).

Storage	Time to finish the record	Capacitor for S2 (10 ms)	for S3 (20 ms)
FRAM ring (record written when made)	56 µs	**353 µF** input side	704 µF
SD only (RAM ring flushed, 3 × 32 KB + 2 file-system writes at the 250 ms busy limit [R14])	1.25 s	49 000 µF, or **0.53 F** supercapacitor	0.53 F

Decision: FRAM ring, 1000 µF/50 V on the input (2.8× margin for ageing and tolerance). The SD card stays as an archive that may lose its last seconds.

8. Analysis of alternatives (W ch. 32, pp. 682–702)

8.1 Sensor set (S6)

Alternatives: A1 one wheel sensor (today's recorder); A2 two wheel sensors; A3 two wheel sensors + IMU; A4 wheel sensor + Doppler radar; A5 two wheel sensors + IMU + radar. Criteria and weights (W Table 32.1, p. 691); the accuracy scores come from S3 (10 at the target, −3 per decade above it); the others are assumptions with the reason below.

Criterion	Weight	A1	A2	A3	A4	A5
brake distance under slide (p95, target 5 cm)	0.30	4.04	4.73	**8.60**	7.79	7.66
last-1500 m distance (p95, target 0.25 m)	0.15	5.99	4.23	6.75	5.51	6.84
unit cost	0.15	10	9	8	4	3
installation and upkeep	0.15	10	9	9	4	4
availability (degraded modes)	0.15	3	6	8	6	9
software assurance effort	0.10	9	8	6	7	4
weighted total		6.46	6.45	**7.94**	5.96	6.12

Reasons for the assumed scores: a MEMS IMU costs a few euros and sits on the recorder board (no extra cabling); a ground radar under a street-running low-floor tram needs a mounting under the car body, cleaning and an aiming calibration [R10 notes the aiming error]; A1 has no second source when its sensor fails; the estimator in A3/A5 is more software to assure.

Robustness (W p. 692 and Fig. 32.4): A3 wins 70.1 % of 20 000 Dirichlet-random weightings (A1 28.8 %, when cost and upkeep get most of the weight); no ±10 % change of any single weight changes the winner.

8.2 Storage: FRAM ring against SD only — §7.4 (D7 in §15).

8.3 The brake estimator (S3, S4)

Option	What it does in the brake	p95 all conditions	Why it is (not) chosen
E3 (chosen)	no wheel updates; IMU integral from v0; boundary correction v(T) = 0; pitch gyro for gravity	**14.7 cm**	best in every condition
E3h	as E3, but the wheel distance when no slide is seen	worse: the braking creep (2 % wet) is not a slide (S3)	rejected (D3)

E3g	wheels used whenever the gate accepts them	73 cm	small slides inside the gate add up
E3a	E3 without the gyro	1.9 m	gravity changes on vertical curves and under body pitch
E5	E3 + radar updates	30 cm	the radar's ±0.5 % scale replaces the boundary information

Why the boundary correction works. During the brake the inertial speed is $v(t) = v0 + \int(f - \hat{c})\,dt$. An offset error δc that stays constant over the brake leaves a speed residual $r = \delta c\, T$ at the standstill, where the true speed is 0. Removing that offset gives $d = D(T) - r\,T/2$ exactly (OR-23). An offset error that grows linearly (gyro offset error δb: $\delta c = g\,\delta b\,t$) leaves $-g\,\delta b\,T^3/12$; that is why the gyro offset is learnt at every stop and tracked between stops.

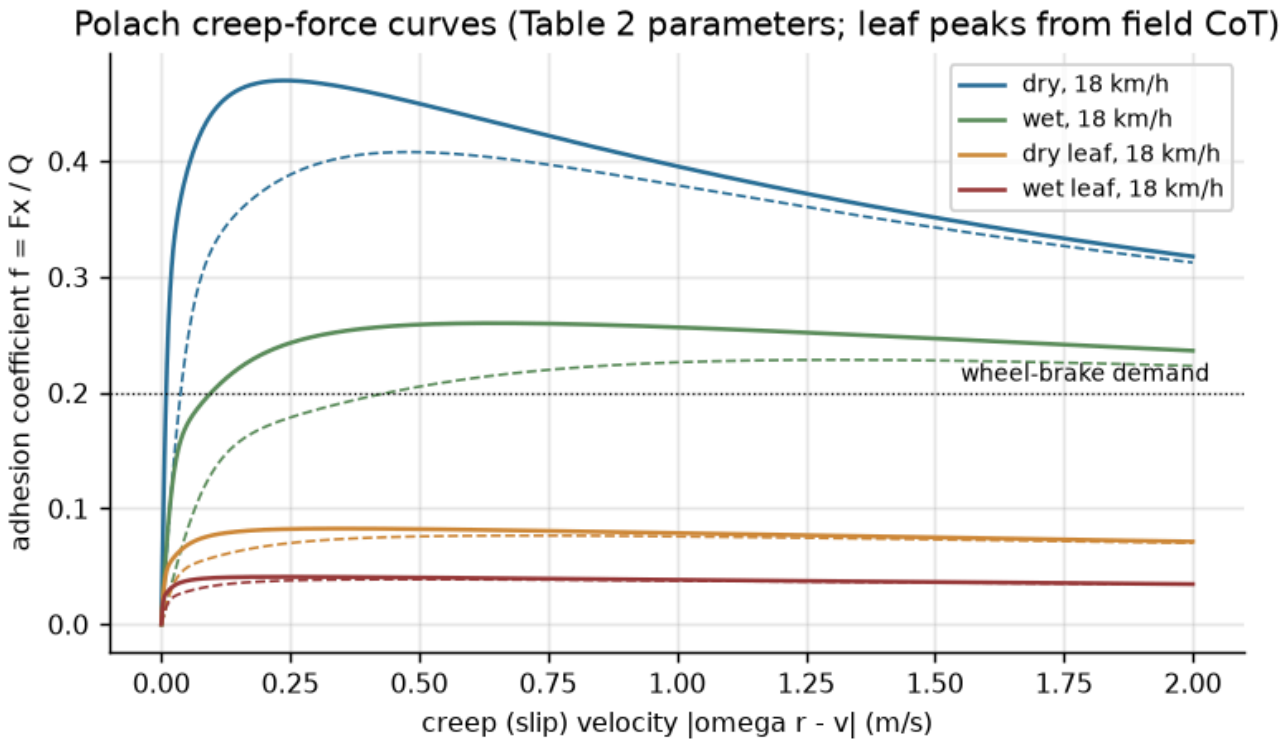
9. Modelling and simulation (W ch. 33, pp. 703–720)

9.1 M&S plan (W p. 707)

Model	Purpose	Fidelity	Validation
Polach creep force [R2] with Hertz contact	wheel-rail force vs creep	Table 2 parameters; leaf peaks scaled to [R3]	test_physics: Table 2 values, leaf peaks, analytic slope
tram brake dynamics: 6 axles, implicit wheel solver, WSP, track brakes, gradients, vertical curves, body pitch (1.5 Hz)	wheel speeds and true motion	2 ms steps in the brake	S1 against BOStrab Table 2
sensor models: quadrature edges with capture times; ADXL355-class accelerometer [R9]; BMI088-class gyro [R20]; radar ±0.5 % [R10]; vibration 1–10 Hz	what the recorder sees	data-sheet errors drawn per run	—
estimators E1–E5 (`model/odo/estimator.py`)	candidates	double precision	synthetic runs with known answers (test_estimator)
firmware `fw/lib/odo`	the product	float32	equals the Python E3 within 2 mm on four recorded scenarios (OR-80)

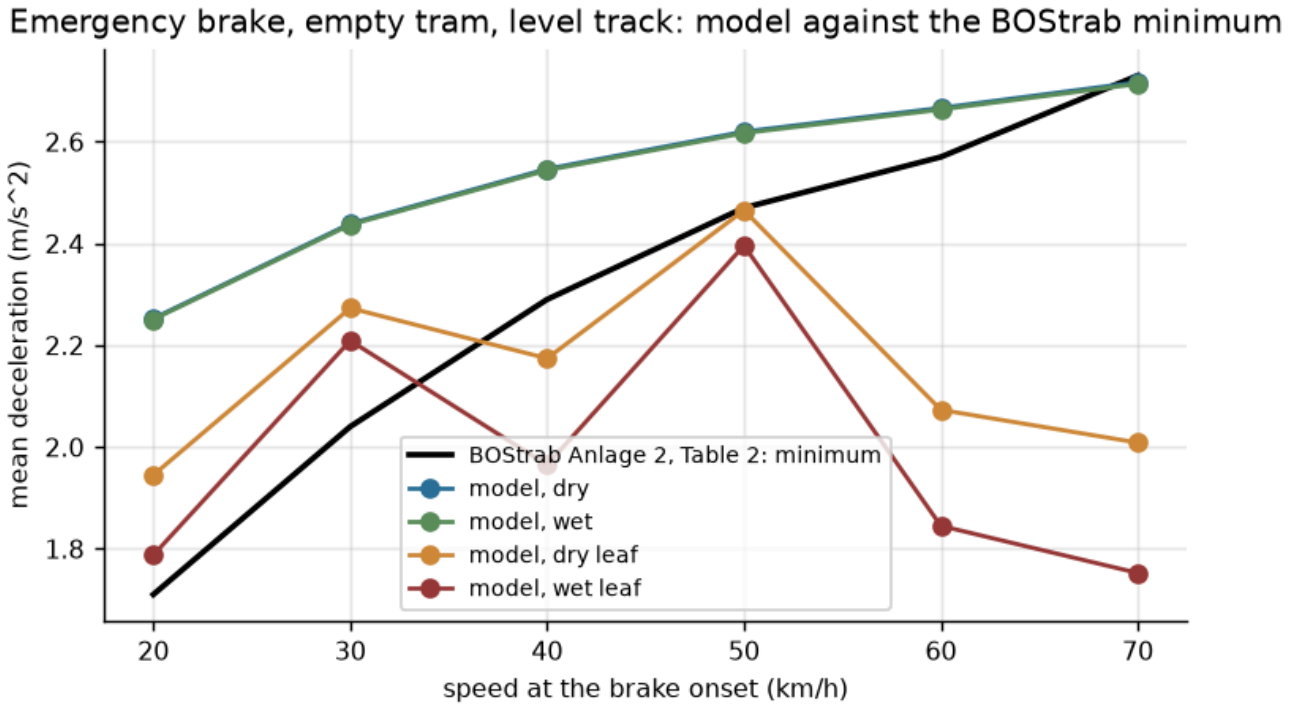
Monte Carlo envelope (S3): 4 rail conditions × 48 runs; speed 30–70 km/h; brake after 1550–1650 m from a 5 s standstill; gradient ±8 % (±5 % with leaves, sanding not modelled); a vertical curve of up to ±4 % change over 40–120 m around the brake; mass 38–52 t; wheel diameters 635–680 mm per bogie; sensor errors drawn from the data-sheet ranges.

9.2 Results: braking physics (S1)

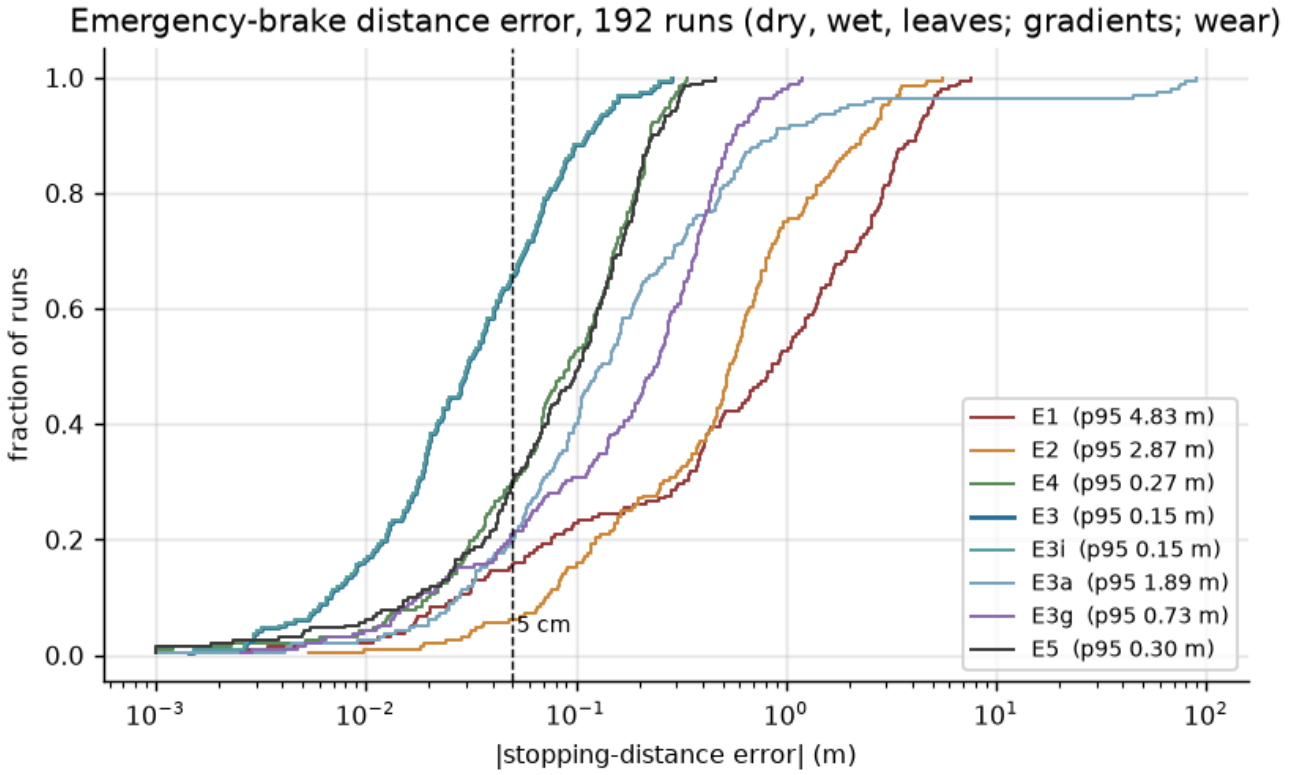


- Peak adhesion at 36 km/h: dry 0.44, wet 0.25, dry leaves 0.08, wet leaves 0.04 (the last two set to [R3]).
- Creep needed for the wheel-brake share (f = 0.2): **0.19 % dry, 2.0 % wet**: the wheels under-read by this much without any slide.

- Empty tram, level track: dry and wet meet BOStrab Table 2 from 20 to 60 km/h (e.g. 2.62 m/s² against 2.47 at 50 km/h) and are 0.4 % short at 70 km/h (2.72 against 2.73) because of the jerk-limited build-up of the wheel brakes (A). With leaves the mean deceleration falls to 1.75 m/s² at 70 km/h (108 m against 69 m), as expected without sanding.



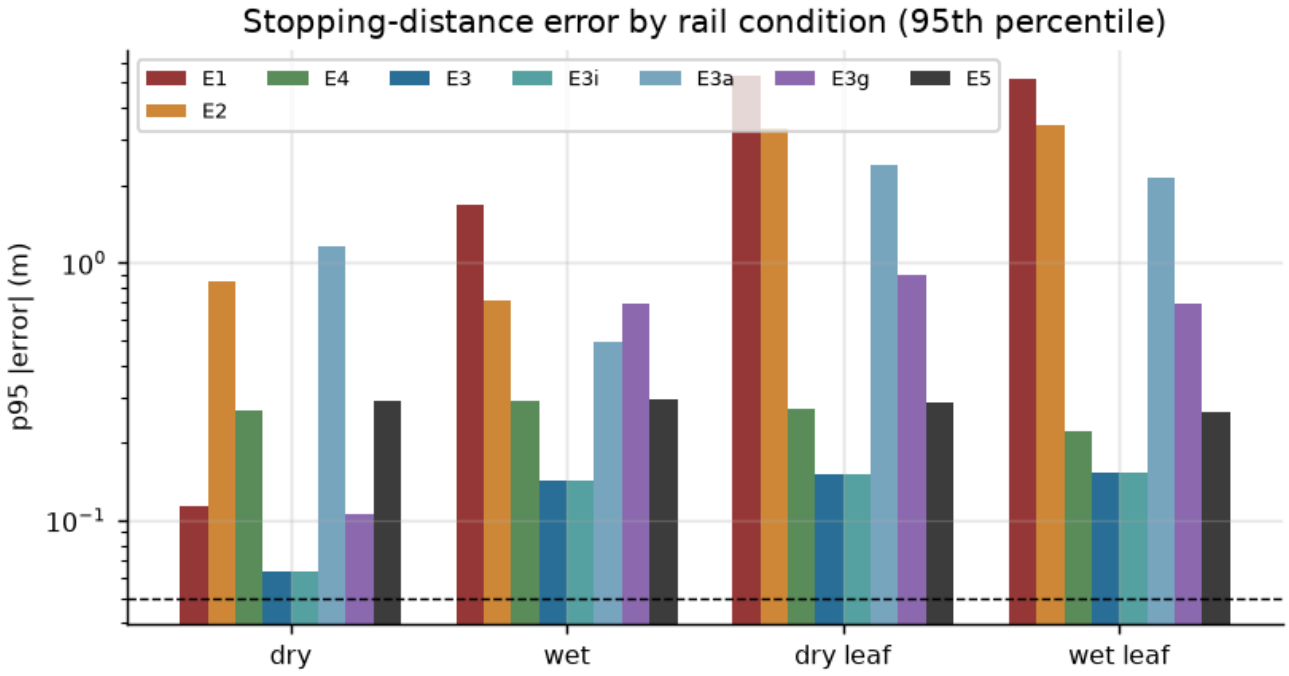
9.3 Results: estimators over the envelope (S3)



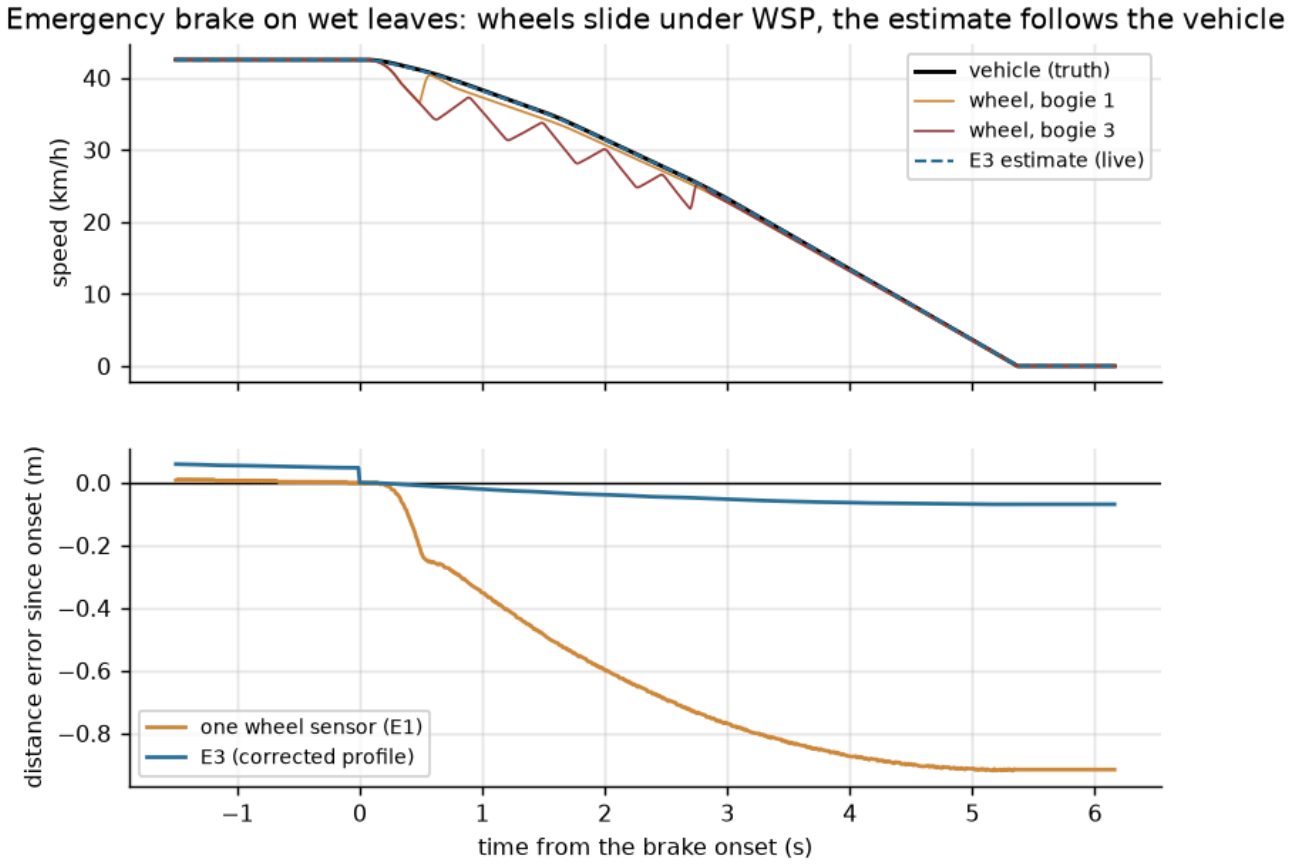
Estimator	p50	p95	worst	inside 5 cm	1500 m window p95
E1 one wheel sensor	0.86 m	4.83 m	7.56 m	16 %	5.4 m
E2 two wheel sensors (larger increment)	0.54 m	2.87 m	5.58 m	6 %	21 m
E4 Doppler radar	9.2 cm	27 cm	34 cm	30 %	7.9 m
E3 wheels + IMU (chosen)	**3.0 cm**	**14.7 cm**	**28.6 cm**	**66 %**	**3.0 m**

E3a without the gyro	13 cm	1.89 m	89 m	20 %	3.9 m
E3g wheels used when gated in	24 cm	73 cm	1.19 m	21 %	3.0 m
E5 E3 + radar	10 cm	30 cm	47 cm	30 %	2.8 m

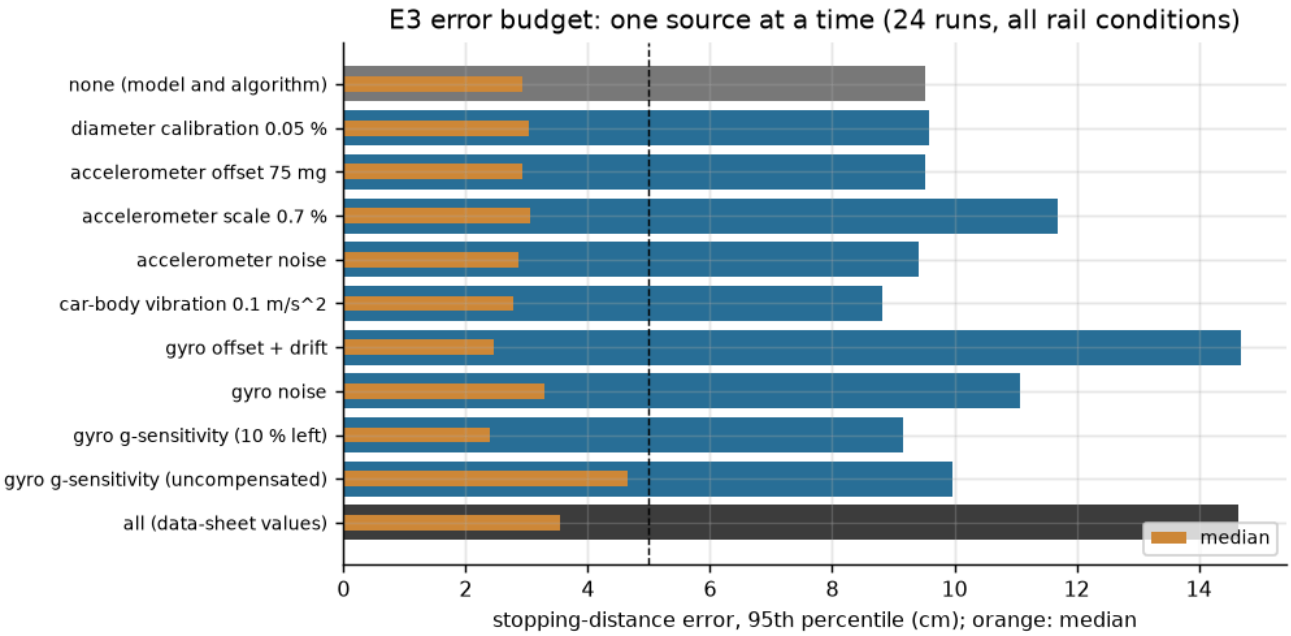
By rail condition (p95): E1 0.11 m dry, 1.69 m wet, 5.3 m dry leaves, 5.2 m wet leaves; E3 6.4 cm dry, 14.3 cm wet, 15.1 cm dry leaves, 15.3 cm wet leaves.

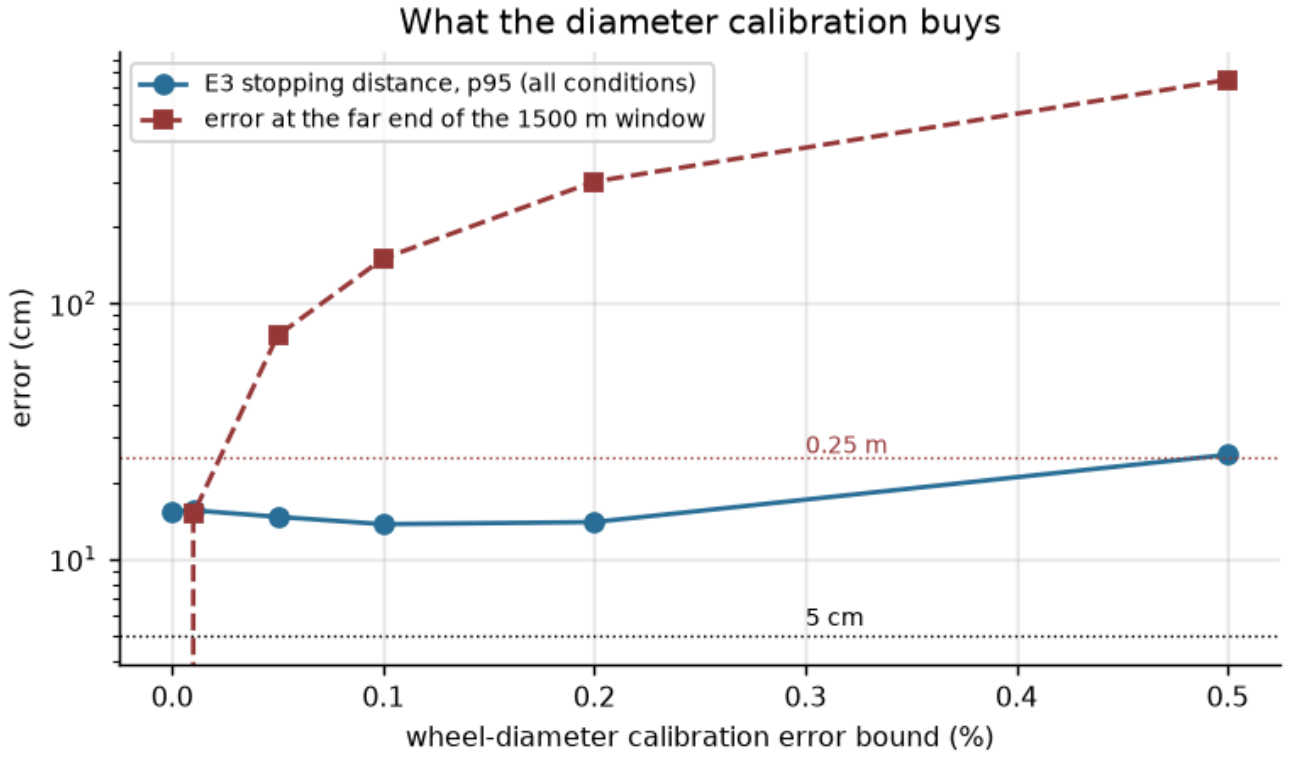


One emergency brake on wet leaves (WSP cycling, both bogies sliding): the one-sensor distance drifts to −2.9 m, the E3 corrected profile stays within 0.1 m.



9.4 Results: error budget and calibration (S4)





Depot brake test (dry, level, empty, 30–50 km/h, stopping distances 14–37 m):

Diameter calibration	E3 p95	E3 inside 5 cm	E1 p95	E1 inside 5 cm
0.01 %	2.5 cm	100 %	3.0 cm	100 %
0.05 %	2.5 cm	100 %	3.5 cm	100 %
0.1 %	3.0 cm	100 %	4.5 cm	100 %

On a dry level track the one-sensor recorder also meets 5 cm if its diameter is right; what it cannot do is any brake with slide or wet-rail creep.

9.5 Results: hold-up (S5)

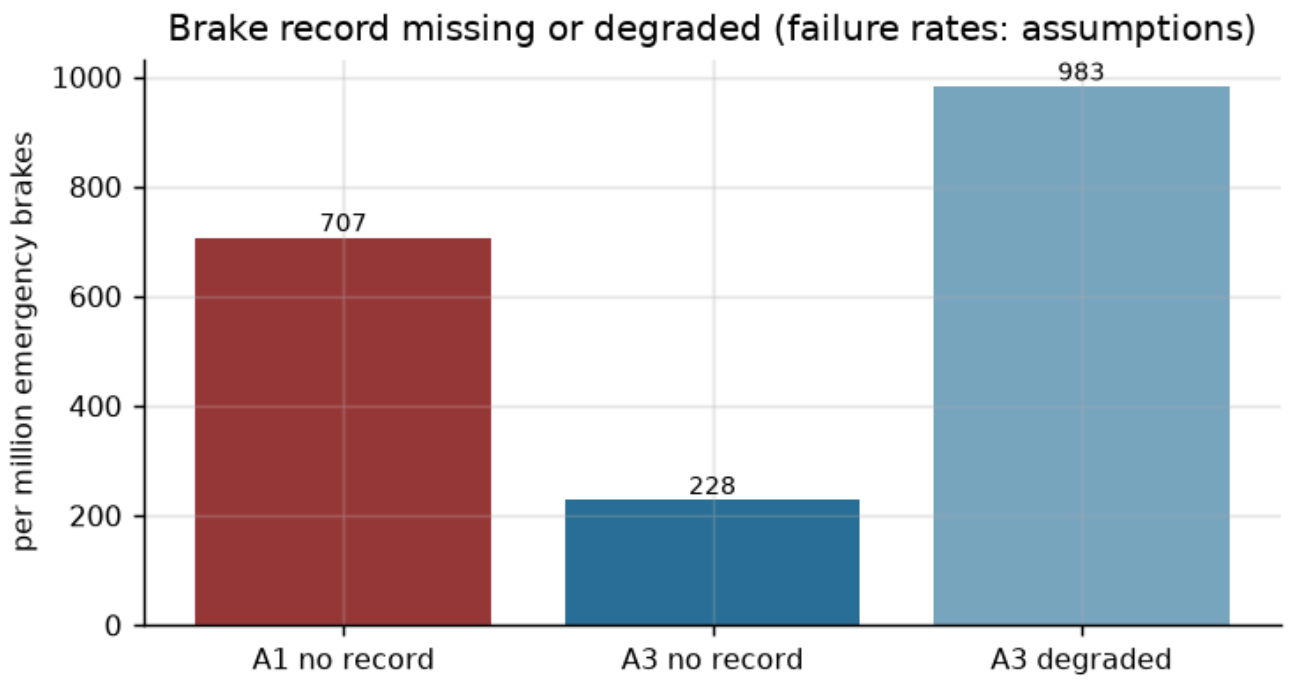


10. Reliability, maintainability, availability (W ch. 34, pp. 721–791)

FMEA (after W Fig. 34.20, p. 766; detection is part of the design, W Principle 34.28, p. 760):

Row	Item / failure mode	Local effect	System effect	Detection	Mitigation
W1	one wheel sensor dead or cable cut	no edges	estimator on the other wheel	OR-17: rest while the other wheel and the IMU move > 1 m/s for 2 s	flag WHEEL0/1; repair at depot
W2	both wheel sensors dead	no standstill evidence	brake event stays open	OR-17 on the first, then no edges at all	flagged; no wrong result is written
W3	wheel slide / spin	wheel speed wrong	none (gated out)	gate OR-14	by design
I1	IMU stuck at full scale	prediction wrong	wheel-only, no slide correction	OR-50 range check	flag IMU; OR-53
I2	IMU offset drift	slow error	learnt at every stop	standstill updates OR-13	daily standstill check
F1	FRAM write cut by the supply loss	one torn slot	none	CRC at recovery	OR-33
F2	FRAM device failure	records lost	no record	boot recovery finds no valid slot	watchdog/BIT (A)
P1	power-fail comparator dead	no PFAIL record	last record may be torn	monthly interruption test (A)	OR-33 keeps all complete records
C1	wrong diameter entered	scale error	1500 m window wrong; brake insensitive	limits OR-41; CALBAD flag OR-42	calibration run
B1	brake-demand input open	no event	brake not recorded as an event	every depot brake test exercises it	test procedure

Quantitative estimate (S7; failure rates are assumptions in the range of field data: wheel sensor 20 000 FIT, board 3000 FIT, IMU 500 FIT, power-fail circuit 400 FIT, FRAM 50 FIT; time in service until repaired 24 h when detected on-line, plus half the test interval when latent):



- Probability that an emergency brake has **no record at all**: 707 per million brakes for A1 (its single wheel sensor is a single point of failure) against 228 per million for A3.
- A3 brakes recorded in a **degraded** mode (IMU down, or one wheel sensor down): 983 per million.
- A1's real weakness is not the missing record but the **wrong** one: every brake with slide or wet-rail creep (about 40 % of the S3 mix) is under-read by metres.

11. Qualification plan

- **EN 50155 [R4]:** supply range 0.7–1.25 Un and 0.6–1.4 Un for 100 ms; interruption class S2 (hold-up per §7.4); temperature class for the cab (–25..+85 °C [R19]); vibration and shock to EN 61373 (not read).
- **EMC:** EN 50121-3-2 (not read). The power-fail comparator needs a filter that ignores bursts but reacts within the 10 ms hold-up.

- **Recorder function:** brake tests on a measured track with a reference (optical correlation sensor or GNSS-RTK), dry and with artificial low adhesion (H1 in OPEN_ITEMS.md).

12. Software assurance: EN 50716 / EN 50657-style plans and evidence

EN 50657 governs on-board rolling-stock software; it was derived from EN 50128 and its SIL1–SIL4 requirements are equivalent to EN 50128:2011; EN 50716:2023 now replaces both [R5] (V2). The SIL of a recorder function comes from the operator's hazard analysis (EN 50126 [R6]); a recorder is usually SIL 0–2 (A). The repository applies the techniques of the higher SILs as a developer would:

Technique	Evidence
software requirements with IDs and rationale	`fw/SRS.md` (33 requirements)
traceability requirement → test, checked	`fw/tools/trace.py` in CI: 33/33
coding standard	MISRA C:2012 via cppcheck: no findings outside one advisory deviation (`fw/docs/misra.md`)
structural coverage	host tests: 100 % statements (596), branches (350), MC/DC condition outcomes (348), GCC 14
functional tests on the target code	Renode IT01–IT08 on the RT1064 image
independent reference	the Python estimator (double precision) against the firmware (float32): within 2 mm
defensive programming	no dynamic memory, bounded loops, CRC on frames and records, plausibility checks

Not done (needs an organisation): independence of verification, tool qualification, assessment (H8).

13. Verification and validation (W ch. 13, pp. 270–292; ch. 28, pp. 599–622)

13.1 Strategy (W Principles 13.3/13.4, p. 275; 13.11, p. 283)

Verification by analysis (S1–S7), by test on the host (unit tests, recorded scenarios), and by test on the emulated target (Renode). Validation: the brake requirement is validated against the BOStrab stopping distances (§5) and the dynamic model against BOStrab Table 2 (S1).

13.2 Integration (W Figs. 28.2/28.4)

Models → recorded scenarios (`tools/export_scenarios.py`) → firmware library on the host (same scenarios) → firmware image on the emulated RT1064 with the rig (same scenarios, plus power, calibration and frame tests).

13.3 Evidence produced in this repository

Check	Result
Model tests (`model/tests`)	11 tests: source values, analytic derivatives, synthetic runs with known answers
Host unit tests (`fw/scripts/test-host.sh`)	471 checks; 100 % lines, branches and MC/DC
Recorded scenarios on the host (OR-80)	depot dry +8 mm; wet 6 % downhill –78 mm; wet-leaf slide +64 mm; sag curve –130 mm (one wheel sensor: –11 mm, –1.05 m, –5.15 m, –3.10 m)
MISRA C:2012	no findings outside D1
Traceability	33/33
Renode IT01–IT08	see README "Firmware evidence"

13.4 Validation

The values that decide the answer (slide, creep, vibration, pitch) are model inputs; OPEN_ITEMS.md lists the vehicle and laboratory measurements that would validate them.

14. What is sourced, what is assumed, and the limits

- **Sourced (V):** BOStrab limits [R1], Polach parameters [R2], leaf adhesion [R3], ADXL355 [R9], BMI088 [R20], radar accuracy [R10], the recorder specification [R19]. **Secondary (V2):** EN 50155 classes [R4], EN 50657/50716 relation [R5], IEC 62625-1 scope [R7], wheel range [R11], gradients [R13], SD busy [R14].
- **Assumed (A):** the tram (mass, axles, track brakes, wheel-brake demand, WSP thresholds), the leaf patch layout, body pitch and vibration, the gyro g-sensitivity calibration, power and converter, failure rates. The assumptions table is in `docs/sources.md`.
- **Model limits.** The wheel-rail model is Polach's empirical curve, not a contact-mechanics solution with third-body layers; sanding is not modelled; the track brakes have a constant force; the vehicle is one rigid mass with pitch as a filtered response.

- **Emulator limits.** Renode's RT1064 has no quadrature decoder and no IMU: the rig sends the sampled front end as frames (I1–I3). The FRAM is an array in RAM. Cycle counts are Renode's virtual time.
- **Not certified.** The process is EN 50716/EN 50657 *style* only.

15. Decision log

ID	Decision	Evidence
D1	Brake distance from the IMU with a two-point boundary correction, not from the wheels	S3: E3 p95 14.7 cm against E1 4.83 m
D2	A pitch gyro is part of the sensor set	S3: E3a (no gyro) p95 1.89 m; vertical curves and body pitch change gravity along the car
D3	No wheel distance in the brake, also without a seen slide (E3h rejected)	S1: braking creep 2 % wet; E3h worse in every condition
D4	A wheel at rest gives no speed measurement; standstill updates only while $\|f - c\| < 0.15$ m/s²	found by the models: at departure the first edges come late and pulled the filter to 0
D5	The brake-demand edge is captured with its time	10 ms = 19 cm at 70 km/h; IT02 inside 5 cm
D6	Standstill = end of the deceleration, confirmed by wheel rest and the inertial speed	locked wheels on leaves stop turning before the tram stops
D7	FRAM record ring; SD only as an archive	S5: 353 µF against 0.53 F
D8	Calibration over two marks 100 m apart at walking pace	mark sampled every 10 ms; IT07 0.003 %
D9	"0.25 m over 1500 m" is implemented as the recording step; the accuracy is the budget of §7.2	S3, S4
D10	IMU fault: the wheels carry the brake without correction	OR-53; the filter cannot follow a brake without the IMU

16. Open engineering questions, and why they matter to a recorder maker

1. Measured creep and slide of a low-floor tram on wet rail and leaves, with a reference speed (H1): the whole case rests on the creep and slide levels.
2. The longitudinal vibration and pitch spectrum at the cab mounting (H2): the vibration term of the budget.
3. Can the gyro offset be learnt during cruise well enough to drop the dependence on recent stops? (The filter already tracks it; a long run without a stop is the open case.)
4. Can a wheel-turning record from the depot system update the diameter automatically and close the 1500 m window budget?

17. References

Wasson

[W] C. S. Wasson, *System Engineering Analysis, Design, and Development: Concepts, Principles, and Practices*, 2nd ed. Hoboken, NJ: Wiley, 2016. Chapters 4–7, 13, 19–22, 26–28 and 31–34; pages in the text.

Braking, adhesion and vehicles

[R1] Verordnung über den Bau und Betrieb der Straßenbahnen (BOStrab), Anlage 2, Tables 1 and 2.

[R2] O. Polach, "Creep forces in simulations of traction vehicles running on adhesion limit," *Wear* 258 (2005) 992–1000.

[R3] J. Jaffé, K. Tomlinson, R. Lewis, "Moisture: the most important factor in leaf low adhesion?," *Tribology – Materials, Surfaces & Interfaces* 20(3) (2026) 189–205.

[R8] EN 15595, *Railway applications – Braking – Wheel slide protection* (concept only, not read).

[R11] UIC leaflet 510-2 (wheel diameters; via a search excerpt).

[R13] Published tram gradients (secondary sources; see docs/sources.md).

Standards for rolling-stock electronics and software

[R4] EN 50155:2021, *Railway applications – Rolling stock – Electronic equipment* (classes via supplier summaries).

[R5] EN 50657:2017+A1:2023 and EN 50716:2023 (relation to EN 50128 via standards-body abstracts).

[R6] EN 50126-1/-2, EN 50129 (background).

[R7] IEC 62625-1:2013, *On board driving data recording system – Part 1: System specification* (abstract).

Sensors and storage

[R9] Analog Devices, ADXL354/ADXL355 data sheet, Rev. A.

[R10] L. Du et al., "Speed calibration and traceability for train-borne 24 GHz continuous-wave Doppler radar sensor," *Sensors* 20(4) (2020) 1230.

[R12] Two-channel Hall speed sensors for rail traction (60–300 pulses/rev; secondary).

[R14] SD card write busy limit 250 ms (SD Physical Layer specification, via driver commits).

[R20] Bosch Sensortec, BMI088 data sheet, rev. 1.9.

Product reference

[R19] A published tram speed/event recorder specification (name withheld under the repository naming rule).

Revision history

Rev	Date	Change
A	11 Nov 2026	First issue